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# Subject: Azure SLA and Service Lifecycles:

## Unit 5 : Introduction

Computer Science & Engineering

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# Outline

- Service Lifecycle in Azure
- Service Lifecycle Stages
- Overview of Planned Maintenance in Azure
- Overview of Updates in Azure
- Deprecation Policies in Azure

# Service Lifecycle in Azure

- In the past, software development was a closed process, and only a few people could use the software until it was officially released.

- Now, companies like Microsoft let people try their products and services early in development. **This helps them get feedback from users, find out what features are needed, and improve how the product works with other systems.**

- In Azure Cloud, this process is known as the service lifecycle. It describes how an Azure service is developed and made available to the public.

- The service lifecycle ensures that new products are well-tested and refined based on user feedback before becoming widely available. This approach helps Microsoft deliver reliable and high-quality services in Azure.

# Service Lifecycle Stages

**Internal Development:**

- The Microsoft product team starts by identifying requirements and building the product through several iterations.

**Private Preview:**

- After initial development, the product is shared with a select group of users for early feedback.
- These users test the product and provide direct feedback to the development team.
- Private Previews come without any service level agreements (SLAs) or general customer support.
- Products in this stage are offered "as is," and not all will progress to the next stage.

# Service Lifecycle Stages Cont...

**Public Preview:**

- When the product is more mature, it becomes available to all Microsoft users in the Public Preview stage.
- These products still have no SLAs and are typically free.
- Public Previews help gather broader feedback to determine if the product meets quality standards.
- There's a risk that products might be discontinued if they don't meet expectations, so using them in critical applications is not recommended.

**General Availability (GA):**

- Products that reach this stage are fully available for use and often come with billing.
- Unlike previews, GA products have SLAs and full customer support.
- These products are stable enough for integration into production workloads

# Overview of Planned Maintenance in Azure

Planned maintenance in Azure is a structured process where Microsoft schedules updates and improvements to its cloud infrastructure**. This maintenance ensures security, performance, and reliability**. Here's a comprehensive look at how Azure handles planned maintenance:

**Notification Process:**

- **Advance Notice:** Azure provides notifications well in advance of planned maintenance. This helps customers prepare and take necessary actions to minimize disruption.
- **Channels of Notification:** Notifications are sent through multiple channels:
  - **Azure Portal:** Notifications are displayed directly in the Azure portal.
  - **Email Alerts:** Emails are sent to the administrators and contacts specified in the Azure subscription.
  - **Service Health Alerts:** Custom **alerts can be configured in Azure Service Health to receive notifications about maintenance events**.

# Overview of Planned Maintenance in Azure Cont..

## Maintenance Windows:

- **Scheduled Windows:** Maintenance activities are scheduled during predefined windows, chosen to minimize impact based on usage patterns. Typically, these windows avoid peak usage times.

- **Customizable Windows:** In some cases, Azure allows customers to specify preferred maintenance windows to align with their business hours and minimize disruption.

# Overview of Planned Maintenance in Azure Cont..

**Types of Maintenance:**

- **Hardware Maintenance:**
  - **Upgrades and Replacements:** Physical components such as servers, storage devices, and networking equipment are upgraded or replaced to maintain performance and reliability.
  - **Impact Mitigation:** Azure uses live migration techniques to move virtual machines (VMs) to updated hardware without downtime whenever possible.
- **Software Maintenance:**
  - **Patches and Updates:** Regular software patches are applied to the underlying operating systems, hypervisors, and management software to ensure security and performance.
  - **Service Improvements:** Updates are made to Azure services to introduce new features, fix bugs, and improve efficiency.

# Overview of Planned Maintenance in Azure Cont..

**Impact on Services:**

- **Live Migration:** Azure often uses live migration to transfer VMs to updated hardware without causing downtime. This technique helps minimize the impact on running services.
- **Reboots:** Some maintenance activities might require a reboot. Azure aims to keep these instances to a minimum and schedules them during low-usage periods.
- **Service Health and Monitoring:** Azure provides real-time status updates through the Azure status page and Service Health, allowing customers to monitor the progress and impact of maintenance activities.

# Overview of Planned Maintenance in Azure Cont..

## Customer Actions:

- **Set Up Alerts:** Configure Azure Service Health alerts to receive notifications about planned maintenance events.
- **High-Availability Features:**
  - **Availability Sets:** Deploy VMs in availability sets to ensure that at least one instance remains running during maintenance.
  - **Availability Zones:** Distribute resources across multiple availability zones to enhance fault tolerance and minimize downtime.
  - **Redundant Resources:** Implement redundant resources and failover strategies to ensure application continuity.
- **Review and Prepare:** Regularly review maintenance notifications and prepare for any necessary adjustments or backups. This might include temporarily scaling down non-critical services or shifting workloads to unaffected regions.

# Overview of Planned Maintenance in Azure Cont..

**Communication Channels:**

- **Azure Service Health:**
  - Provides detailed information about planned maintenance events, including timelines, affected services, and potential impacts.
  - Allows customers to customize alert rules to stay informed about issues that specifically affect their resources.
- **Azure Status Page:**
  - Offers a real-time overview of Azure services' health globally.
  - Displays information about ongoing and upcoming maintenance activities, outages, and other service-impacting events.

# Overview of Updates in Azure

- **Regular Update Schedule**:
    1. Azure services are updated regularly, including both scheduled and unscheduled updates. Major updates may introduce new features, while minor updates may focus on security patches or bug fixes.
- **Notification of Updates**:
    1. **Azure Portal**: Users are notified of updates through the Azure portal. Notifications include details about the nature of the update, its impact, and scheduled times.
    2. **Email and Communication Channels**: Important updates are communicated via email to registered users. Azure Service Health alerts can also provide real-time notifications.
    3. **Release Notes**: Azure publishes release notes and update logs on their official website, detailing all changes, new features, improvements, and fixes.

# Overview of Updates in Azure Cont...

- **Types of Updates**:
  1. **Security Updates**: Critical patches to address vulnerabilities and enhance the security of Azure services.
  2. **Feature Updates**: New functionalities and enhancements added to existing services to improve capabilities.
  3. **Performance Improvements**: Optimizations aimed at enhancing the speed, reliability, and efficiency of Azure services.
  4. **Bug Fixes**: Corrections to known issues to improve stability and user experience.

# Overview of Updates in Azure Cont...

- **Deployment of Updates**:
  1. **Phased Deployment**: Updates are often rolled out in phases to reduce the risk of widespread issues. This allows Azure to monitor the impact and address any problems quickly.
  2. **Automatic Updates**: Users can choose to enable automatic updates for certain services, ensuring they always have the latest features and security enhancements without manual intervention.
  3. **Manual Updates**: Users have the option to apply updates manually, providing control over the timing of updates to avoid disruption during critical operations.

# Overview of Updates in Azure Cont...

**1.Impact on Services**:

1. **Minimal Downtime**: Azure aims to minimize downtime and service disruption during updates. Most updates are applied with zero or minimal interruption to services.
2. **Advance Notice**: For updates requiring service interruptions, Azure provides advance notice, allowing users to plan accordingly.
3. **Service Health Dashboard**: Users can check the Azure Service Health dashboard for real-time information about ongoing and upcoming maintenance activities.

**2.Testing and Feedback**:

1. **Pre-Release Testing**: Updates undergo rigorous testing in various environments to ensure compatibility and performance. This includes internal testing, as well as beta testing with select users.
2. **User Feedback**: After updates are released, Azure encourages users to provide feedback to help identify any issues and improve future updates. Feedback can be submitted through the Azure portal or via support channels.

# Overview of Updates in Azure Cont…

- **Rollback and Support**:
  1. **Rollback Options**: In case an update causes significant issues, Azure provides support for rolling back to previous versions or configurations.
  2. **Technical Support**: Azure's support team is available to assist with any problems arising from updates. Users can access support through the Azure portal, support tickets, or phone support.
  3. **Documentation**: Comprehensive documentation is available to guide users through update processes, troubleshooting, and rollback procedures.
- **Compliance and Documentation**:
  1. **Compliance Standards**: Updates ensure that Azure services comply with industry standards and regulations, such as GDPR, HIPAA, and ISO certifications.
  2. **Detailed Documentation**: Azure provides extensive documentation, including update logs, implementation guides, and best practice recommendations to help users understand and apply updates effectively.

# Overview of Updates in Azure Cont...

- **Monitoring and Reporting**:
  1. **Update Monitoring**: Users can monitor the status and impact of updates via the Azure portal. This includes information about update progress, affected services, and any issues encountered.
  2. **Reporting Tools**: Azure offers reporting tools that allow users to track update history, performance metrics, and compliance status. These tools help in maintaining transparency and accountability.
- **Best Practices**:
  1. **Regular Review**: Users are encouraged to regularly review update notifications and apply updates promptly to ensure their services remain secure and up-to-date.
  2. **Testing in Non-Production Environments**: Before applying updates to critical workloads, it is recommended to test them in a non-production environment to identify potential issues.
  3. **Backup and Recovery Plans**: Having a robust backup and recovery plan in place ensures that data and services can be restored quickly in case of any issues during updates.
  4. **Stay Informed**: Users should stay informed about upcoming updates, new features, and best practices by following Azure's official blogs, forums, and documentation.

# Deprecation Policies in Azure

Users may handle the move away from deprecated services with little disturbance and uninterrupted operational efficiency by adhering to Azure's deprecation regulations and best practices.

- **Understanding Deprecation**:
  1. **Definition**: Deprecation in Azure refers to the phasing out of features, services, or APIs that will no longer be supported or available in future releases.
  2. **Impact**: Deprecated services may eventually become unavailable, requiring users to migrate to newer alternatives.
- **Deprecation Announcement**:
  1. **Advance Notice**: Azure provides ample advance notice before deprecating any service or feature. Typically, users are notified 12 to 24 months in advance.
  2. **Communication Channels**: Deprecation announcements are made through multiple channels, including the Azure portal, email notifications, Azure blogs, and official documentation.
  3. **Documentation**: Detailed documentation and guidance are provided, explaining the reasons for deprecation, the timeline, and the recommended alternatives or migration paths.

# Deprecation Policies in Azure Cont..

- **Deprecation Phases**:
  1. **Announcement Phase**: Azure announces the deprecation, providing details on the timeline, impact, and necessary actions for users.
  2. **Transition Phase**: Users are given a specified period to transition from the deprecated service to a supported alternative. During this phase, both the deprecated service and its replacement are available.
  3. **End of Life (EOL) Phase**: The deprecated service is fully retired and becomes unavailable. Users must have completed their migration by this time.
- **Migration Support**:
  1. **Guidance and Tools**: Azure provides detailed migration guides, best practices, and tools to assist users in transitioning to alternative services.
  2. **Support Channels**: Dedicated support channels are available to help users with migration-related queries and issues.
  3. **Compatibility Information**: Information on compatibility and differences between the deprecated service and its replacement is provided to facilitate a smooth transition.

# Deprecation Policies in Azure Cont..

- **Impact on SLAs and Pricing**:
  1. **Service Level Agreements (SLAs)**: Once a service is marked for deprecation, Azure ensures that SLA commitments are honored until the end of the deprecation period. Post-EOL, SLA commitments no longer apply.
  2. **Pricing Adjustments**: Pricing for deprecated services may remain unchanged during the deprecation period. Users are informed of any pricing changes related to the alternative services.
- **Examples of Deprecation**:
  1. **API Deprecation**: Older API versions may be deprecated in favor of newer, more efficient versions. Users are provided with updated API documentation and migration guides.
  2. **Service Deprecation**: Entire services may be deprecated if they are outdated or replaced by more advanced offerings. For example, Azure Managed Disks replacing Unmanaged Disks.
  3. **Feature Deprecation**: Specific features within a service may be deprecated while the service itself remains active. Users are guided on how to adapt their usage.

# Deprecation Policies in Azure Cont..

- **Best Practices for Handling Deprecation**:
  1. **Stay Informed**: Regularly check Azure's announcements, blogs, and documentation to stay informed about upcoming deprecations.
  2. **Early Planning**: Begin planning and executing migrations as soon as a deprecation is announced to avoid last-minute issues.
  3. **Utilize Azure Support**: Leverage Azure's support and migration tools to ensure a smooth transition.
  4. **Test New Solutions**: Thoroughly test the alternative services in a non-production environment before fully migrating to ensure compatibility and performance.
  5. **Monitor Updates**: Keep an eye on updates and enhancements to the replacement services, as they may offer additional benefits and features.
- **Deprecation Examples**:
  1. **Azure AD Graph API**: Deprecated in favor of Microsoft Graph API. Users were provided with migration guides to transition their applications.
  2. **Azure Storage Analytics**: Deprecated in favor of Azure Monitor. Comprehensive documentation and migration strategies were shared with users.
  3. **Classic Azure Virtual Machines**: Deprecated in favor of Azure Resource Manager (ARM) based VMs. Users received detailed instructions on how to migrate their resources.

# References:

1. Cheshire, J. (2020). Exam Ref AZ-900 Microsoft Azure Fundamentals, 2nd Edition. Microsoft Press.
2. Miles S. (2021). Microsoft Azure Fundamentals Certification and Beyond, Packt Publishing.

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